WEEK 4: COMPARISON OF BPE, WORDPIECE AND SENTENCEPIECE TOKENIZERS

1. Aim

To implement and compare three subword tokenization approaches, namely Byte Pair Encoding (BPE), WordPiece and SentencePiece, using English, Telugu and Hindi text and observe how each tokenizer splits text into tokens and generates token IDs.

2. Software Requirements

- Python 3.10 or later
- Visual Studio Code
- Python Virtual Environment
- Internet Connection

3. Libraries Used

1. **Transformers:** Used to load pretrained tokenizers from Hugging Face.
2. **SentencePiece:** Used to support SentencePiece tokenization.
3. **AutoTokenizer:** Used to load different pretrained tokenizers automatically.

4. Installation Commands

Install the required libraries using the following commands:

```
pip install transformers
pip install sentencepiece
```

Alternatively, create a `requirements.txt` file containing:

```
transformers
sentencepiece
```

Install the dependencies using:

```
pip install -r requirements.txt
```

5. Project Structure

Create a folder named `Week4` in Visual Studio Code.

Create the following files:

1. tokenizer_comparison.py
2. requirements.txt

6. Tokenization Approaches

A. Byte Pair Encoding (BPE)

BPE is a subword tokenization technique that repeatedly merges frequently occurring adjacent character or symbol pairs to create larger tokens.

Model Used: GPT-2.

B. WordPiece

WordPiece is a subword tokenization technique that uses a learned vocabulary and splits unknown or rare words into smaller meaningful pieces.

Model Used: Multilingual BERT.

C. SentencePiece

SentencePiece is a language-independent tokenization framework that learns subword units directly from raw text.

Model Used: XLM-RoBERTa.

7. Python Program

Create a file named `tokenizer_comparison.py`.

```
from transformers import AutoTokenizer

TOKENIZERS = {
    "BPE - GPT-2": "gpt2",
    "WordPiece - Multilingual BERT": "bert-base-multilingual-cased",
    "SentencePiece - XLM-R": "xlm-roberta-base"
}

TEXTS = {
    "English": "Artificial intelligence is changing the world.",
    "Telugu": "కృత్రిమ మేధస్సు ప్రపంచాన్ని మారుస్తోంది.",
    "Hindi": "कृत्रिम बुद्धिमत्ता दुनिया को बदल रही है।"
}

def load_tokenizers():
    loaded = {}

    for display_name, model_name in TOKENIZERS.items():
        print(f"Loading {display_name}...")
        loaded[display_name] = AutoTokenizer.from_pretrained(
            model_name
        )

    return loaded


def compare_tokenizers(loaded_tokenizers):

    for language, text in TEXTS.items():

        print("\n" + "=" * 75)
        print(f"LANGUAGE: {language}")
        print(f"INPUT TEXT: {text}")
        print("=" * 75)

        for tokenizer_name, tokenizer in loaded_tokenizers.items():

            encoded = tokenizer(
```

```
            text,
            add_special_tokens=True,
            truncation=False
        )

        token_ids = encoded["input_ids"]

        tokens = tokenizer.convert_ids_to_tokens(token_ids)

        print(f"\n{tokenizer_name}")
        print("Tokens     :", tokens)
        print("Token IDs  :", token_ids)
        print("Token count:", len(token_ids))


def main():
    tokenizers = load_tokenizers()
    compare_tokenizers(tokenizers)


if __name__ == "__main__":
    main()
```

8. Execution

Run the following command in the VS Code terminal:

```
python tokenizer_comparison.py
```

The program loads three pretrained tokenizers and processes the English, Telugu and Hindi sentences.

It displays:

- Token pieces
- Token IDs
- Total token count

9. Output

```
[transformers] PyTorch was not found. Models won't be available and only tokenizers, configuration and file/data utilities can be used.
Loading BPE - GPT-2...
Loading WordPiece - Multilingual BERT...
Loading SentencePiece - XLM-R...
\n================================================================================
LANGUAGE: English
INPUT TEXT: Artificial intelligence is changing the world.
================================================================================
\nBPE - GPT-2
Tokens     : ['Art', 'ificial', 'Ġintelligence', 'Ġis', 'Ġchanging', 'Ġthe', 'Ġworld', '.']
Token IDs  : [8001, 9542, 4430, 318, 5609, 262, 995, 13]
Token count: 8
\nWordPiece - Multilingual BERT
Tokens     : ['[CLS]', 'Arti', '##ficial', 'intelligence', 'is', 'changing', 'the', 'world', '.', '[SEP]']
Token IDs  : [101, 90841, 69229, 30151, 10124, 43068, 10105, 11356, 119, 102]
Token count: 10
\nSentencePiece - XLM-R
Tokens     : ['<s>', '_Artificial', '_', 'intelligence', '_is', '_changing', '_the', '_world', '.', '</s>']
Token IDs  : [0, 231996, 6, 130687, 83, 151134, 70, 8999, 5, 2]
Token count: 10
\n================================================================================
LANGUAGE: Telugu
INPUT TEXT: కృత్రిమ మేధస్సు మారుతోంది ప్రపంచాన్ని
================================================================================
\nBPE - GPT-2
Tokens     : ['à', '°', 'ķ', 'à', '±', 'ĥ', 'à', '°', '¤', 'à', '±', 'į', 'à', '°', '°', 'à', '°', 'ç', 'à', '°', '°', 'Ġ', 'à', '°', '°', 'à', '±', 'ĩ
°', '§', 'à', '°', '.', 'à', '±', 'į', 'à', '°', '.', 'à', '±', 'ģ', 'Ġ', 'à', '°', '¾', 'à', '°', '°', 'à', '°', '¾', 'à', '°', 'Ĥ', 'à
ļ', 'à', '°', '¥', 'à', '°', '"', 'à', '±', 'į', 'à', '°', '"', 'à', '°', 'ç', 'Ġ', 'à', '°', '°', 'à', '°', '¾', 'à', '°', '°', 'à', '±', 'ģ', 'à', '°
à', '±', 'į', 'à', '°', '¤', 'à', '±', 'ĩ', 'à', '°', 'Ĥ', 'à', '°', '¦', 'à', '°', 'ç', '.']
Token IDs  : [156, 108, 243, 156, 109, 225, 156, 108, 97, 156, 109, 235, 156, 108, 108, 156, 108, 123, 156, 108, 106, 220, 156, 108, 106, 156, 109, 229,
8, 100, 156, 108, 116, 156, 109, 235, 156, 108, 116, 156, 109, 223, 220, 156, 108, 103, 156, 109, 235, 156, 108, 108, 156, 108, 103, 156, 108, 224, 156,
8, 156, 108, 122, 156, 108, 101, 156, 109, 235, 156, 108, 101, 156, 108, 123, 220, 156, 108, 106, 156, 108, 122, 156, 108, 108, 156, 109, 223, 156, 108,
6, 109, 235, 156, 108, 97, 156, 109, 233, 156, 108, 224, 156, 108, 99, 156, 108, 123, 13]
Token count: 112
\nWordPiece - Multilingual BERT
Tokens     : ['[CLS]', 'క', '##ృ', '##త్రి', '##మ', 'మ', '##ే', '##ధ', '##స్సు', 'మారు', '##తోం', 'ది', '##ప్ర', '##పం', '##చా', '##న్ని', '.', '[SEP]']
Token IDs  : [101, 1201, 111339, 88046, 24696, 1224, 17729, 99288, 15370, 39179, 94504, 23769, 1224, 27374, 15370, 68450, 119, 102]
Token count: 18
\nSentencePiece - XLM-R
Tokens     : ['<s>', '_', 'కృత్', 'రిమ', 'మ', '_మే', 'ధ', 'స్సు', '_మారు', 'తోం', '_ప్రపం', 'చాన్ని', '.', '</s>']
Token IDs  : [0, 6, 151424, 91445, 3071, 22290, 11600, 129461, 86457, 6258, 169610, 104528, 5, 2]
Token count: 14
\n================================================================================
LANGUAGE: Hindi
INPUT TEXT: वृत्रिम बुद्धिमत्ता दुनिया को बदल रही है।
```

```
\n================================================================================
LANGUAGE: Hindi
INPUT TEXT: वृत्रिम बुद्धिमत्ता दुनिया को बदल रही है।
================================================================================
\nBPE - GPT-2
Tokens     : ['à¤', 'ķ', 'à¥', 'ĥ', 'à¤', '¤', 'à¥', 'į', 'à¤', '°', 'à¤', 'ç', 'à¤', '©', 'Ġà¤', '¬', 'à¥', 'ģ', 'à¤', '¦', 'à¥', 'į', 'à¤', '§', 'à¤',
¤', '©', 'à¤', '¤', 'à¥', 'į', 'à¤', '¤', 'à¤¾', 'Ġà¤', '¦', 'à¥', 'ģ', 'à¤', '"', 'à¤', 'ç', 'à¤', '¯', 'à¤¾', 'Ġà¤', 'ķ', 'à¥', 'ĩ', 'Ġà¤', '¬', 'à¤',
¤', '²', 'Ġà¤', '°', 'à¤', '¹', 'à¥', 'Ģ', 'Ġà¤', '¹', 'à¥', 'Ī', 'à¥', '¤']
Token IDs  : [11976, 243, 24231, 225, 11976, 97, 24231, 235, 11976, 108, 11976, 123, 11976, 106, 28225, 105, 24231, 223, 11976, 99, 24231, 235, 11976, 1
6, 123, 11976, 106, 11976, 97, 24231, 235, 11976, 97, 48077, 28225, 99, 24231, 223, 11976, 101, 11976, 123, 11976, 107, 48077, 28225, 243, 24231, 233, 2
5, 11976, 99, 11976, 110, 28225, 108, 11976, 117, 24231, 222, 28225, 117, 24231, 230, 24231, 97]
Token count: 68
\nWordPiece - Multilingual BERT
Tokens     : ['[CLS]', 'व', '##ृ', '##त्', '##र', '##िम', 'ब', '##ुद्ध', '##िम', '##त्ता', 'दुनिया', 'को', 'ब', '##दल', 'रही', 'है', '।', '[SEP]']
Token IDs  : [101, 865, 37500, 18321, 50419, 887, 48010, 50419, 94034, 83874, 11267, 887, 99989, 57203, 10569, 920, 102]
Token count: 17
\nSentencePiece - XLM-R
Tokens     : ['<s>', '_वृत्त', 'रि', 'म', '_बुद्धि', 'म', 'त्ता', '_दुनिया', '_को', '_बदल', '_रही', '_है', '।', '</s>']
Token IDs  : [0, 74707, 97885, 1920, 105202, 1920, 54297, 29405, 629, 14512, 8906, 460, 125, 2]
Token count: 14
```

10. Observations

1. Successfully installed the required libraries, including Transformers and SentencePiece.
2. Implemented three tokenization approaches: BPE, WordPiece and SentencePiece.
3. Loaded pretrained GPT-2, Multilingual BERT and XLM-R tokenizers.

4. Observed how English, Telugu and Hindi sentences were divided into smaller units called tokens.
5. Noticed that different tokenizers generated different token pieces and token counts for the same input sentence.
6. Observed differences in how multilingual and primarily English-trained tokenizers handled different languages.
7. Successfully generated token IDs and token counts for all three languages.
8. Understood that tokenization converts text into numerical representations that can be processed by language models.

11. Result

Successfully implemented and compared BPE, WordPiece and SentencePiece tokenization approaches using English, Telugu and Hindi text. Observed the differences in token splitting, token IDs and token counts across the three tokenizers.